Design and Development of a USB-Powered Smart Cabinet Security System Using Arduino with RFID and Tamper Detection

Abstract

This project presents the design and development of a USB-powered smart cabinet security system using an Arduino microcontroller, RFID authentication, and tamper detection. The system enhances cabinet security by allowing access only to authorized RFID card holders while operating from a standard 5 V USB power source, making it energy-efficient and portable. When a valid RFID tag is detected, the Arduino controls an electronic locking mechanism to unlock the cabinet for a predefined period. Unauthorized RFID cards are denied access, and repeated failed attempts can trigger an alert. A tamper detection sensor continuously monitors the cabinet for forced opening or physical interference. If tampering is detected, the system activates an audible buzzer and visual LED indication to warn users of a potential security breach. The proposed system is low-cost, reliable, and easy to implement using readily available hardware components. It is suitable for securing personal lockers, office cabinets, laboratory storage, and small asset protection applications. Future enhancements may include wireless notifications, cloud-based monitoring, and biometric authentication for improved security and remote access management.